Thermodynamic Chess

A chess engine as an annealing computation: emergent temperature, Boltzmann-counting optionality, and the phase structure of positions

Conceptual summary of the `fablethermochess` engine, second edition (the annealed model) · July 2026

Abstract. Conventional engines value a chess position by minimax. This engine replaces that rule with statistical mechanics, and — in this second edition — removes every prescribed thermodynamic quantity from the model. Energy is geometric material alone. Entropy is optionality: at the search horizon a side's move ensemble is uniform, so its entropy is Boltzmann's own $S = \ln W$ with W the number of available moves. Temperature is not chosen but *measured*: the search is an explicit annealing process in which iterative deepening is time, and the bath temperature is the ensemble-weighted residual fluctuation of the engine's own value revisions — cooling toward the $T = 0$ minimax ground state as the search converges, floored only by the zero-point quantum of the evaluation lattice (one square of reach, derived from the geometry). Position value is free energy $F = \langle Q\rangle + T\cdot S$ at the bath temperature, so activity can trade against material only at the rate the engine's genuine uncertainty licenses. The previous edition's prescribed heat capacity C, its material schedule, and its user slider are gone; the phase structure they exposed survives as pure measurement — every position has an intrinsic capacity curve $C(T) = \mathrm{Var}_\pi(Q)/T^2$ with a Schottky peak C* at spinodal temperature $\hat{T}$, and the dashboard reports where the measured bath sits on it: frozen, cold, critical, or hot. The redesign was driven by adversarial evidence: matches against a pure-material alpha-beta oracle exposed each hidden constant as a blunder source, and the final model plays soundly where its predecessor hemorrhaged material.

1. The central idea

A chess position with the player to move is treated as a physical system. Its accessible microstates are its legal moves; each move *a* has a value Q_a — the negated energy of the microstate (**Q = −E**: good moves are low-energy states) — obtained recursively from the position it leads to. The system occupies its microstates with canonical probabilities and is valued by its free energy:

$$\pi_a = e^{Q_a/T} / Z \,, \qquad F = T \ln Z = \langle Q\rangle + T\cdot S \,, \qquad S = -\Sigma_a\, \pi_a \ln \pi_a \tag{1}$$

Maximizing F is minimizing E − TS. The decomposition is the heart of the design: a position is valuable for the quality of its best continuations *and* for the entropy of having many comparably good ones — but the entropy is paid at the going temperature, and everything hinges on what sets T.

Search proceeds in negamax form, $Q_a = -F(\text{position after a})$, with the identical rule for both sides. As $T \to 0$ the log-sum-exp collapses to the maximum and the engine is exact minimax; the engine always *plays* $\mathrm{argmax}_a\, Q_a$, with the weights π_a displayed as a measurement of how contested the choice was. Equation (1) is the soft Bellman backup of maximum-entropy reinforcement learning, and two-sided Boltzmann best response is a logit quantal-response equilibrium; what distinguishes this engine is that T is an emergent, measured quantity (§4).

2. Energy: geometric material, and nothing else

Nothing is imported from chess folklore — no 1/3/3/5/9 table, no piece-square bonuses. A piece type's value is its mean x-ray reach: the number of squares it sweeps from a square on an empty board, averaged over the 64 squares. The pawn's base is 2 (its typical move count; also the display unit: 1♙ = 2 internal units) and the king's is 0 (never traded; both sides always have exactly one, so any base cancels in the difference). The static energy of a position is the material difference of these bases — *only* that:

	Pawn	Knight	Bishop	Rook	Queen
mean x-ray reach = base	(2)	5.25	8.75	14	22.75
emergent value (♙)	1	2.6	4.4	7	11.4
classical value (♙)	1	3	3	5	9

> **What was removed, and why.** The first edition added each piece's instantaneous blocker-aware mobility to the energy at full weight, making one square of reach worth half a pawn — a hidden hardcoded exchange rate between material and activity, roughly ten times larger than any calibrated engine uses. Match diagnosis against a pure-material oracle showed exactly the failure this predicts: moves that lost a knight and moves that kept material backed up with *identical* values, because routine mobility swings are the same magnitude as a minor piece's base. Mobility is real value, but it is not energy; it is optionality, and optionality is entropy (§3) — priced at temperature, as physics demands. When the engine's uncertainty is low, activity is nearly worthless against material; exactly as it should be.

3. Entropy: optionality as Boltzmann counting

Interior nodes get optionality for free — it is the T·S term of (1), propagated by the backup. Leaves need a static term, and here the model makes its cleanest statement. At the search horizon the engine has *no energy resolution* among the unsearched continuations: the correct ensemble over a side's moves is therefore uniform — maximum entropy — and its entropy is Boltzmann's counting formula $S = \ln W$, with W the side's bare (pseudo-legal) move count. The leaf evaluation is

$$U = \text{material} + T\cdot(\ln W_{us} - \ln W_{them}) \qquad (2)$$

— side-symmetric by construction (no parity bias between odd and even search depths), bounded, logarithmic (the tenth extra move is worth far less than the second), and computable in a single board scan. A position in check contributes material only: passing is illegal there, the ensemble is not stationary, and a non-equilibrium state has no entropy to sell. Trapped pieces, open files, and centralization all still register — as counting, at temperature — but a piece of material can never be outbid by activity unless the bath is hot enough to license it, and the bath is only hot when the engine genuinely cannot tell its best moves apart (§4).

4. Temperature: the annealing residual

There is no prescribed heat capacity, no material schedule, no slider. The search is an explicit annealing computation: iterative deepening is time, deepening is cooling, and the bath temperature is *measured*. Whenever the search revises a root value by looking deeper, the revision $|\Delta Q|$ is one sample of the engine's residual uncertainty about its own evaluations. The bath temperature for the next iteration is the σ-scale of that fluctuation ensemble,

$$T \leftarrow \sqrt{(T_{meas}^2 + T_0^2)}\ , \quad T_{meas} = 1.2533 \cdot \langle\, |Q^{(d)} - Q^{(d-2)}|\, \rangle_\pi \qquad (3)$$

with three qualifications, each learned from a failure mode observed in play:

1. **Ensemble-weighted.** The average in (3) is Boltzmann-weighted by the current occupancies π. The revision of a move the distribution does not occupy is resolved knowledge, not residual fluctuation — an unweighted thermometer reads every refuted blunder as heat and boils.
2. **Same-parity.** Negamax-softmax values carry an alternating entropy ladder — every node adds $+T\cdot S$ for its mover — so values at adjacent depths differ by one uncompensated layer $\sim T\cdot\bar{S}$. A thermometer comparing depth d to $d-1$ reads the ladder itself as fluctuation and feeds back $T \propto T$: exponential runaway heating (observed: $T = 1.4 \rightarrow 3.4 \rightarrow 10.9 \rightarrow 59.7$ within one game). Samples therefore compare depth d with $d-2$ — same leaf parity, ladder cancelled identically. Depth 1 is compared against the static material baseline, so the first reading is the tactical volatility that quiescence uncovered; depth 2 has no clean partner and takes no reading.
3. **Zero-point floor.** The evaluation spectrum is quantized — reach comes in whole squares — and a quantized system's fluctuation cannot fall below its level spacing. T_0 is the minimal positive level spacing of the mobility spectrum, *computed from the geometry tables* (it evaluates to one square). This is the zero-point motion of the evaluation lattice: a converged search cannot distinguish values finer than one square of reach, so optionality retains exactly that much purchase and no more. Without the floor, quiet positions freeze completely and the engine drifts planlessly into repetition; with it, activity is always worth something — about half a pawn of purchase at most.

T is held constant within one deepening iteration — one bath, one temperature, and the transposition table stays coherent — and updated (with damping) between iterations. Forced lines converge and freeze toward minimax; unresolved middlegames run warm; longer thinking time means a colder engine. Perfect play is the fully annealed $T \rightarrow 0$ ground state. The measured T is global for the whole search tree: a genuinely canonical ensemble, in contrast to the first edition's per-node temperatures, which scaled with each node's own value spread and thus made the engine *most* reckless exactly where the stakes were highest.

5. The phase structure, kept as pure measurement

The first edition's central discovery survives with its arbitrariness removed. Every position has an intrinsic heat-capacity curve — the Schottky anomaly of its move ensemble:

$$C(T) = \mathrm{Var}_\pi(Q) / T^2\,, \qquad C^* = \max_T C(T) \text{ at } T = \hat{T} \tag{4}$$

C(T) vanishes at both ends (the distribution freezes onto the best move as $T \rightarrow 0$; σ saturates as $T \rightarrow \infty$) and peaks between. C^* — the largest capacity the position's ensemble can sustain — is a parameter-free invariant of the position, and $\hat{T}$ is its spinodal. Previously C was *demanded* and T solved for, which forced a choice of branch and a pinning rule; now the bath temperature arrives from the annealing measurement and the dashboard simply reports where it sits on the curve: FROZEN (forced positions), COLD (ordered side, $T < \hat{T}$ — the engine resolves the position sharper than its intrinsic scale), CRITICAL ($T \approx \hat{T}$), or HOT (disordered side). The peak search uses a robust median scale of the bulk of the Q distribution, so a mate outlier reads as frozen rather than defining a fake high-temperature branch.

6. Absorbing states: mate and repetition

Checkmate is a ground state, not an energy level. No entropy may attach to a terminal outcome: whenever the best move is a forced win (or every move a forced loss) the backup is pure max Q, with no $T \ln Z$ term —

otherwise "many ways to mate" would outscore the mate itself and the engine would shuffle checks forever (an observed pathology, now a regression test). Mate scores carry distance: each negation ticks them one unit toward zero, so mate-in-2 strictly outranks mate-in-4 and the scores stay node-relative, keeping transposition entries valid across paths. Mate-scored moves are also excluded from the thermometer ensemble (3): absorbing states carry distance ticks, not thermal fluctuation.

Repetition is an absorbing draw, $F = 0$. Any position recurring on the search path scores zero, and the search is seeded with the hashes of every played position since the last irreversible move, so the engine avoids repetition when ahead and steers into it when lost — in match play it repeatedly rescued lost games this way. Zero is exact by the antisymmetry of (2).

7. Quiescence as relaxation; truncation as bath-width

Thermodynamic state variables belong to (locally) equilibrated systems, and a position with hanging pieces is mid-collision. Captures, promotions, and check evasions are relaxed deterministically — alpha-beta at $T = 0$ with MVV-LVA ordering and delta pruning whose margin includes the optionality swing one capture can cause ($\sim T \cdot \Delta \ln W$) — and only quiet positions enter an ensemble.

In the main search, alpha-beta cutoffs are unsound (every child contributes to Z) but contributions decay exponentially: at depth ≥ 2 a shallow ranking pass orders the children, and only those within 3T of the best (capped at 16) are fully recursed; the tail keeps its shallow values as its ~1% share of Z. The truncation width is set by the *bath*: cold searches narrow toward best-line probing exactly where minimax is exact, hot searches keep many live branches. Two children are always fully resolved regardless of temperature — a one-level system yields no fluctuation sample (the thermometer would only ever confirm itself and freeze), and verifying the runner-up is precisely what catches a shallow misranking of the best move.

8. Chemical potential: pieces, and the tempo as the (N+1)-th piece

With p_P the Boltzmann weight of all moves originating from piece P, freezing P in place costs free energy

$$\mu_P = -T \ln(1 - p_P) \tag{5}$$

(the ideal-gas/independent-pieces limit). A piece with $\mu_P \approx 0$ is thermodynamically idle whatever its material worth; the piece carrying most of the ensemble weight is what the position is *about*.

8.1 The tempo

Apply (5) to the *tempo* — the piece whose moves are all of them — and the formula seems to diverge ($p = 1$) until the ensemble is extended by the null move as a microstate, with energy Q_{pass} equal to the relaxed value of passing. On $Z' = Z + e^{Q_{pass}/T}$, deleting the tempo's moves leaves exactly the pass, formula (5) applies verbatim, and it closes to

$$\mu_{tempo} = -T \ln \pi'_{pass} = T \cdot \ln(1 + e^{\tilde{\chi}/T}) , \qquad \tilde{\chi} \equiv F - Q_{pass} \tag{6}$$

— a softplus, at the bath temperature, of the raw susceptibility reading $\tilde{\chi}$ (§9). Every limit is chess: when the tempo genuinely matters ($\tilde{\chi} \gg T$), $\mu_{tempo} \to \tilde{\chi}$; in zugzwang ($\tilde{\chi} < 0$) it goes smoothly to zero — the tempo of a player in zugzwang is a worthless particle, and no special case is needed; the crossover width is T itself. One formula now prices rook, queen, pawn, *and initiative*, and the dashboard's μ panel carries a tempo row accordingly.

9. Prophylaxis as a response function: the grand potential

King safety and prophylaxis resisted every static formulation we tried: whether a threat is sound is not a counting fact but a search fact, and each attempted leaf term (attack maps, filtered check counts) was the standard hand-crafted evaluator being rederived in disguise. The model's own division of labor points elsewhere: measure it the way physics measures stiffness — perturb and relax. For each candidate move, the *tempo-susceptibility probe* gives the opponent their reply followed by a free tempo (a null move for the mover) and relaxes the result through quiescence, in which unsound sacrifices refute themselves. The reading $\tilde{\chi} = Q - V_{taxed}$ feeds (6), and each candidate acquires a tempo chemical potential.

The opponent is a *tempo reservoir*: their forcing moves annihilate the mover's tempi, with occupancy w measured as the fraction of quiet quiescence-entry positions whose stand-pat is refuted by a forcing line (in-check entries count as taxed). The evaluation is then the grand potential of the position under tempo exchange:

$$\Phi = F - w \cdot (\mu_{tempo} - \bar{\mu}) \tag{7}$$

with $\bar{\mu}$ the minimum over candidates. The subtraction is gauge fixing, not pragmatism: chemical potentials are defined only up to a reference, and only μ *differences* drive exchange — the stiffest candidate pays nothing, and the universal price of one tempo (common mode) cancels. The linear-in-w form is the quenched average over the reservoir's action, the correct choice for an adversary who chooses rather than samples. Absorbing states are exempt as always: a found mate owns its tempo and is never taxed. Three residues, named: Q_{pass} is relaxed at $T = 0$ (quiescence), so the probe inherits quiescence's blindness to quiet mating nets; only the dominant Boltzmann set is probed, and selection stays within it; χ is orthogonal information — negamax at any depth only ever evaluates trajectories where every threat is answered, so the forced-pass counterfactual appears in no ordinary search line at any depth.

10. The experimental record

The redesign was driven by adversarial testing against two opponents: a deliberately naive alpha-beta engine whose evaluation is *pure classical material* (a material-truth oracle that punishes hung pieces and values nothing else), and the first-edition engine itself. Equal time per move, color-swapped pairs, small opening book for game diversity. The oracle exposed each hidden constant in turn:

- **First edition vs oracle: lost 0–2 (+2 escapes), 22 blunders of ≥2.5♙.** Decomposition of the blunder positions showed material-losing and material-keeping moves backing up with equal values (the 1:1 mobility-energy rate), and temperatures of 7–9 in collapsing positions (per-node T scaling with the value spread).
- **Material-only energy with root-measured T: still lost.** The thermometer read refuted moves as heat (fixed by ensemble weighting), read the entropy ladder as heat and ran away exponentially (fixed by same-parity sampling), and froze in quiet positions into planless repetition (fixed by the zero-point floor). Each fix is now load-bearing physics in §4.
- **Final model vs oracle (12 games): won the match, +2 −1 =9.** The engine holds material against a dedicated material engine (comparable blunder counts, 25 vs 20, where the first edition committed 22 in four games)

and converts activity into the winning margin — the intended division of labor between energy and entropy. Most games are draws, as they should be between a material-sound engine and an engine that values only material.

- **Final model vs first edition (12 games): 10–1 with 1 draw**, committing fewer than half the material blunders (23 vs 56).
- **The susceptibility probe, as first implemented, failed decisively (1.5/12 vs the baseline's 6.5/12)** — and the autopsy vindicates the framework's own rules. The probe's counterfactual world (opponent reply plus a free tempo, relaxed only through quiescence) declares mate-scale outcomes in routine middlegames; those readings entered the grand-potential tax unguarded, so a shallow, noisy mate-detector overruled the deep search at measured weight $w \approx 0.7$, with a feedback loop (losing → more forcing positions → higher w → heavier taxes). The violated principle is the model's oldest: absorbing states are not thermal — the probed move's Q was exempted, but the probe's own reading was not. The mechanism ships off by default; μ_{tempo} remains on the dashboard as a measurement. Candidate repairs, untested: saturate probe readings at the bulk scale, and let a low-resolution instrument break only the ties the high-resolution one left unresolved (within T of the best).

That a sequence of purely *physical* repairs — ensemble averaging, parity-clean thermometry, zero-point motion — was what turned a blundering engine into a sound one is itself the most encouraging evidence that the thermodynamic frame is doing real work here, rather than decorating a chess engine with vocabulary.

11. The measurement apparatus

The dashboard reports, per position: the material energies U_w, U_b; the measured bath temperature T and root entropy S; ⟨Q⟩, TS, and F; the Schottky readings $C_{eff} = C(T_{bath})$ and C* with the phase label; the move ensemble (Q_a, π_a); the chemical potentials; and time series of Eval, T, S, TS, C_{eff}, and C* across the game. Four dedicated instruments visualize the physics directly:

- **Capacity curve.** The position's full Schottky anomaly C(T) on a log-temperature axis, with the peak (C^*, $\hat{T}$) marked and the measured bath temperature drawn as a phase-colored marker — where the marker sits on the curve *is* the phase.
- **Annealing trace.** The cooling schedule of the search that just ran: bath T per deepening iteration, with the zero-point quantum T_0 shown as the floor it rides on.
- **Phase strip.** A colored band under the history graph giving the game's phase geography move by move — frozen / cold / critical / hot.
- **μ overlay.** An optional glow on the board itself, ring intensity proportional to each piece's chemical potential: a direct picture of which pieces carry the position's free energy.

The headline Eval is everywhere the argmax-Q value in White-POV pawns — the energetic part alone, comparable across engines — while F carries the optionality term on its own row. Stockfish runs in a separate worker as an external reference thermometer and never influences play; the Δ(engine−SF) column measures where geometric-thermodynamic valuation diverges from conventional valuation. There is no temperature knob anywhere in the interface: T is a reading, not a setting. The one control that remains — thinking time — is physically meaningful: it is the annealing schedule, and a longer think is a colder engine.

12. What this might tell us about chess

- **T(game) is now a real observable.** The bath temperature is the engine's measured self-uncertainty. Its trajectory across a game — where it spikes, how it decays with thinking time, how it correlates with human "sharpness" — is data about chess, not about a tuning choice.
- **C*(position) is a parameter-free invariant**: the largest evaluation-fluctuation capacity a position's move ensemble can sustain. Mapping C* over master games (openings vs technique phases, calm vs sharp players) is a concrete experiment this instrument can run.
- **The entropy ladder is a theorem about soft game values.** Boltzmann backups in zero-sum trees carry an alternating $T\cdot\bar{S}$ layer per ply; any analysis comparing soft values across depths (or any annealed self-play scheme) must sample at matched parity or read the ladder as signal. Discovered here as an engine bug; true of soft minimax generally.
- **Optionality has a measurable exchange rate.** With energy purely material and entropy purely counting, the value of activity in units of pawns is $T\cdot\Delta\ln W$ — and T is measured. In this engine's games activity is worth of order half a pawn when uncertain and nearly nothing when resolved; whether human play implies a similar rate is an open, quantitative question.

Appendix: equation → implementation map

Object	Where it lives
Boltzmann weights, F backup (eq. 1)	`computeF()`, `analyzeRoot()`
Geometric material bases (§2)	`xrayMobility()`, `avgMobility()`, `PIECE_BASE`, `fast_raw_mat()`
$S = \ln W$ leaf optionality (eq. 2)	`staticEval()`, `fast_mob_counts()`
Annealing thermometer (eq. 3)	`annealRoot()` (weighted same-parity samples), `bathT`, `measT`
Zero-point quantum T_0	`T_ZERO` (derived from the mobility spectrum)
Schottky curve, C*, $\hat{T}$, phase (eq. 4)	`measureSchottky()`, `_sigmaOverT()`
Absorbing mates, distance ticking (§6)	`tickMate()`, mate guards in `computeF()`/`analyzeRoot()`
Repetition seeding from the played game (§6)	`collectPastKeys()` → `PAST_KEYS`
Quiescence relaxation, T-aware margin (§7)	`quiesce()`
Bath-width truncation, top-2 rule (§7)	`thermoSearch()`, `BOLTZ_SIGMA`, `TRUNC_MAX`
Chemical potential μ_P (eq. 5)	`computeChemicalPotentials()`, μ panel
Reference thermometer (§10)	`initStockfish()` et seq. (never influences play)